

SSY345 Sensor Fusion HA 1

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1 Properties of random variables

1.1 (a)

i)

Given $z = Aq$ where A is a constant matrix, we use the integral definition of the expectation:

$$\mathbb{E}[z] = \int zp(q)dq = \int (Aq)p(q)dq \quad (1)$$

Since A is a constant matrix, it can be moved outside the integral:

$$\mathbb{E}[z] = A \int qp(q)dq = A\mathbb{E}[q] \quad (2)$$

Thus, we have shown that $\mathbb{E}[z] = A\mathbb{E}[q]$.

ii)

Using the result from (i), the covariance of z is:

$$Cov[z] = \mathbb{E}[(z - \mathbb{E}[z])(z - \mathbb{E}[z])^T] \quad (3)$$

$$= \mathbb{E}[(Aq - A\mathbb{E}[q])(Aq - A\mathbb{E}[q])^T] \quad (4)$$

$$= \mathbb{E}[A(q - \mathbb{E}[q])(q - \mathbb{E}[q])^T A^T] \quad (5)$$

Pulling the constant matrices A and A^T out of the expectation:

$$Cov[z] = A\mathbb{E}[(q - \mathbb{E}[q])(q - \mathbb{E}[q])^T]A^T = ACov[q]A^T \quad (6)$$

This confirms that $Cov[z] = ACov[q]A^T$.

1.2 (b)

Using the derivations from 1a), we calculate the mean and covariance of z :

• **Mean:** $\mathbb{E}[z] = A\mathbb{E}[q] = \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} = \begin{bmatrix} -3 \\ 10 \end{bmatrix}$.

• **Covariance:** $Cov[z] = ACov[q]A^T = \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -0.5 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 \\ -4 & 8 \end{bmatrix}$.

The figure1 illustrates the linear transformation of a random variable q into a new variable z using the matrix A .

The center (mean) of the distribution has moved horizontally. While the mean of q is at $x_1 = 2$, the mean of z shifted to the left at $x_1 = -3$.

The blue ellipse (q) is vertical, meaning its components are uncorrelated. However, the yellow ellipse (z) is tilted toward the top-left, which shows that z now has a negative correlation.

This "tilting" and shifting effect is caused by the structure of matrix A . Specifically, the value -0.5 in the matrix creates a dependency where an increase in x_2 leads to a decrease in x_1 , resulting in the diagonal shape you see.

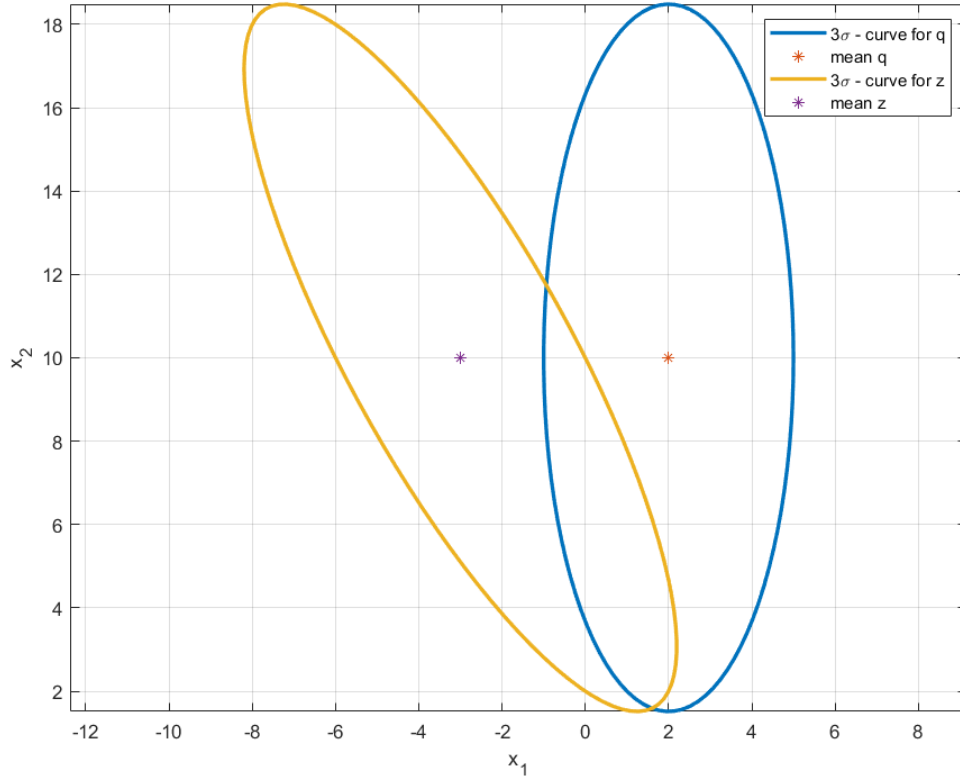


Figure 1: Mean and covariance ellipses for q and z .

2 Transformation of random variables

2.1 (a)

Given $x \sim \mathcal{N}(0, 1)$ and the linear transformation $z = 5x + 3$. The mean and variance are calculated as follows:

$$\mathbb{E}[z] = \mathbb{E}[5x + 3] = 5\mathbb{E}[x] + 3 = 5(0) + 3 = 3. \quad (7)$$

$$\text{Var}[z] = \text{Var}[5x + 3] = 5^2 \text{Var}[x] = 25(1) = 25. \quad (8)$$

Since a linear transformation of a Gaussian variable remains Gaussian, the resulting density is $p(z) = \mathcal{N}(z; 3, 25)$.

The figure 2 demonstrates that for the linear transformation, the resulting probability density remains Gaussian. There is a perfect overlap between the histogram of transformed samples, the analytical PDF, and the numerical Gaussian approximation.

This confirms that linear transformations only shift and scale the distribution without distorting its fundamental symmetric shape. Additionally, the high number of samples used in the simulation ensures that the histogram is smooth and the numerical results converge accurately to the theoretical values.

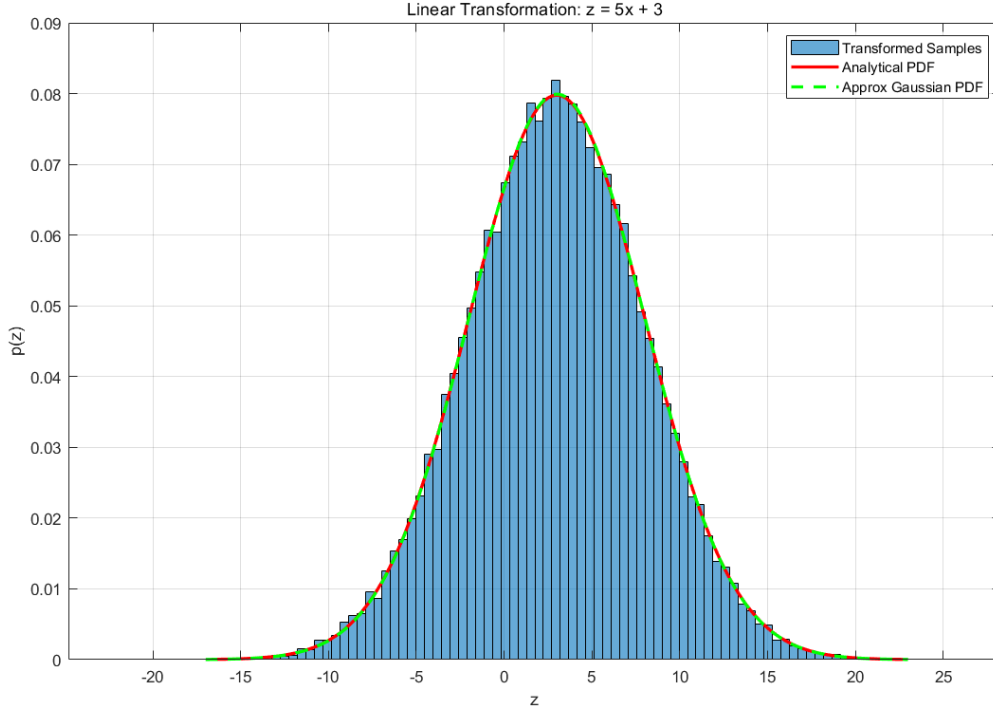


Figure 2: Linear Transformation.

2.2 (b)

Now consider $z = e^x$. While the exact distribution of z is Log-normal, the assignment asks to evaluate the Gaussian approximation[cite: 34].

The analytical mean is:

$$z = e^x, \mathbb{E}[z] = e^{\mu_x + \sigma_x^2/2} = e^{0+1/2} \approx 1.6487 \quad (9)$$

The analytical Variance is:

$$Var[z] = (e^{\sigma_x^2} - 1)e^{2\mu_x + \sigma_x^2} = (e^1 - 1)e^1 \approx 4.6708 \quad (10)$$

The figure 3 that the non-linear transformation $z = e^x$ results in a probability distribution that is non-Gaussian and right-skewed.

While the Gaussian approximation (red line) correctly utilizes the calculated mean and variance, it fails to capture the true asymmetry and the distinct probability peak of the underlying density.

Furthermore, the Gaussian model severely underestimates the "tail risk," as it ignores the significant probability of extreme values clearly visible in the true samples.

Finally, the smoothness of the histogram indicates that a large sample size was employed to ensure the numerical results accurately reflect the non-linear statistics.

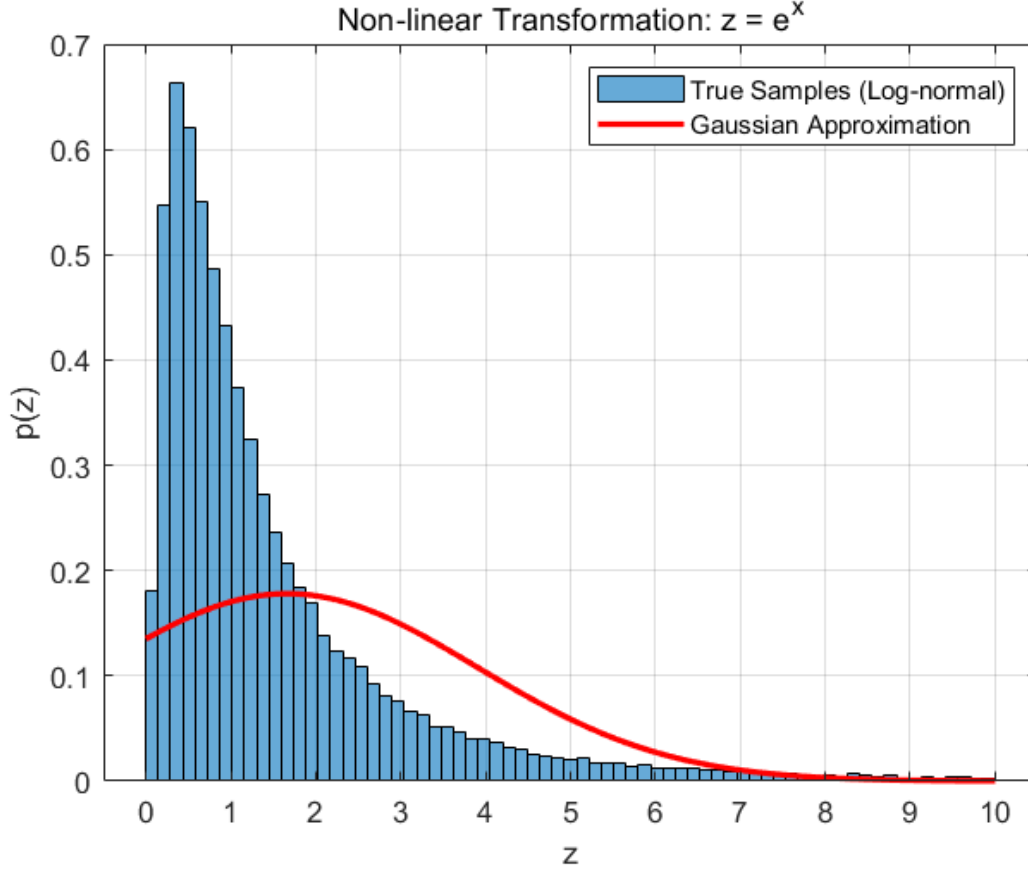


Figure 3: Non-linear Transformation $z = e^x$

2.3 (c)

Comparing the results, we conclude that linearity is the deciding factor for preserving the Gaussian nature of a random variable. While linear models allow for exact statistical descriptions using only the first two moments (mean and variance), non-linear transformations distort the distribution shape, making simple Gaussian approximations insufficient for capturing the true characteristics of the state uncertainty.

3 Understanding the conditional density

3.1 (a)

No, it is not possible to fully describe $p(y)$ with the given information. While we know $y = h(x) + r$, the distribution of y depends on the distribution of x , which has not been specified yet. Without knowing whether x is Gaussian, Uniform, or another type, $p(y)$ remains undefined.

3.2 (b)

Yes, it is possible to describe the conditional density $p(y|x)$. When x is given, the deterministic function $h(x)$ evaluates to a specific constant value. Since $y = h(x) + r$ and the noise r is distributed as $\mathcal{N}(0, \sigma_r^2)$, adding the constant $h(x)$ simply shifts the mean of the

distribution. Therefore, the conditional distribution of y given x is:

$$p(y|x) = \mathcal{N}(y; h(x), \sigma_r^2) \quad (11)$$

This holds true regardless of whether $h(x)$ is linear or non-linear, as long as it is deterministic.

3.3 (c)

When $h(x) = Hx$, the transformation is linear but $p(y)$ remains unknown as long as $p(x)$ is not specified. If x is uniform, y will be non-Gaussian. However, $p(y|x)$ is always Gaussian because the linear term Hx becomes a constant once x is given.

3.4 (d)

When x is Gaussian:

1. If $h(x)$ is linear, $p(y)$ is Gaussian ($\mathcal{N}(H\mu_x, H^2\sigma_x^2 + \sigma_r^2)$).
2. If $h(x)$ is non-linear, $p(y)$ is non-Gaussian (similar to the e^x case in Q2).
3. In both cases, $p(y|x)$ remains Gaussian $\mathcal{N}(h(x), \sigma_r^2)$.

3.5 (e)

The top figure 4 shows when x follows a uniform distribution, the resulting marginal density $p(y)$ is a non-Gaussian, rectangular-shaped distribution (a convolution of uniform and Gaussian). This confirms that $p(y)$ cannot be analytically described as Gaussian unless $p(x)$ is specifically known to be Gaussian.

The below figure 4 is opposite. The conditional density $p(y|x = 2)$ follows a perfect Gaussian distribution, even though $h(x)$ is non-linear. This is because conditioning on x treats $h(x)$ as a deterministic constant, leaving only the Gaussian noise r as the source of uncertainty. This visual proof validates that $p(y|x) = \mathcal{N}(h(x), \sigma_r^2)$.

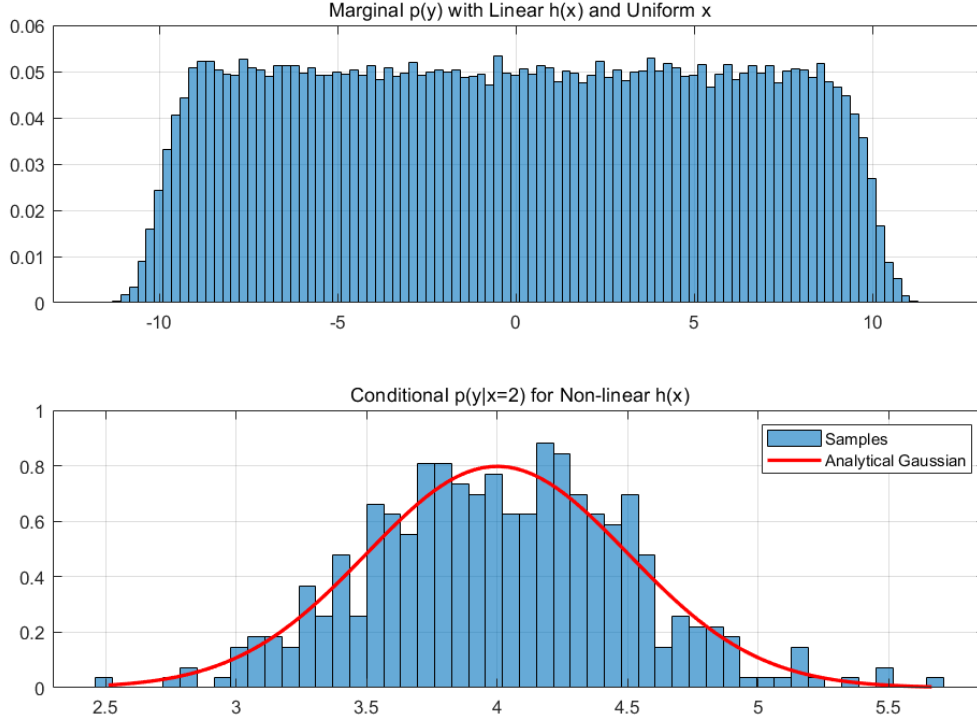


Figure 4: Marginal and Conditional Probability Distributions

4 MMSE and MAP estimators

4.1 (a)

The distribution is looked like in figure 5 with two distinct components. This is a bimodal distribution.

The distribution of y is the result of the convolution between the discrete prior $p(\theta)$ and the continuous Gaussian noise $p(w)$.

Since $p(\theta) = 0.5\delta(\theta - 1) + 0.5\delta(\theta + 1)$, the resulting density is the sum of two shifted Gaussians:

$$p(y) = 0.5\mathcal{N}(y; -1, \sigma^2) + 0.5\mathcal{N}(y; 1, \sigma^2) \quad (12)$$

Because the standard deviation ($\sigma = 0.5$) is small relative to the distance between the two possible values of θ (distance is 2), the two Gaussian "humps" remain largely separated, creating the two visible peaks centered at -1 and $+1$.

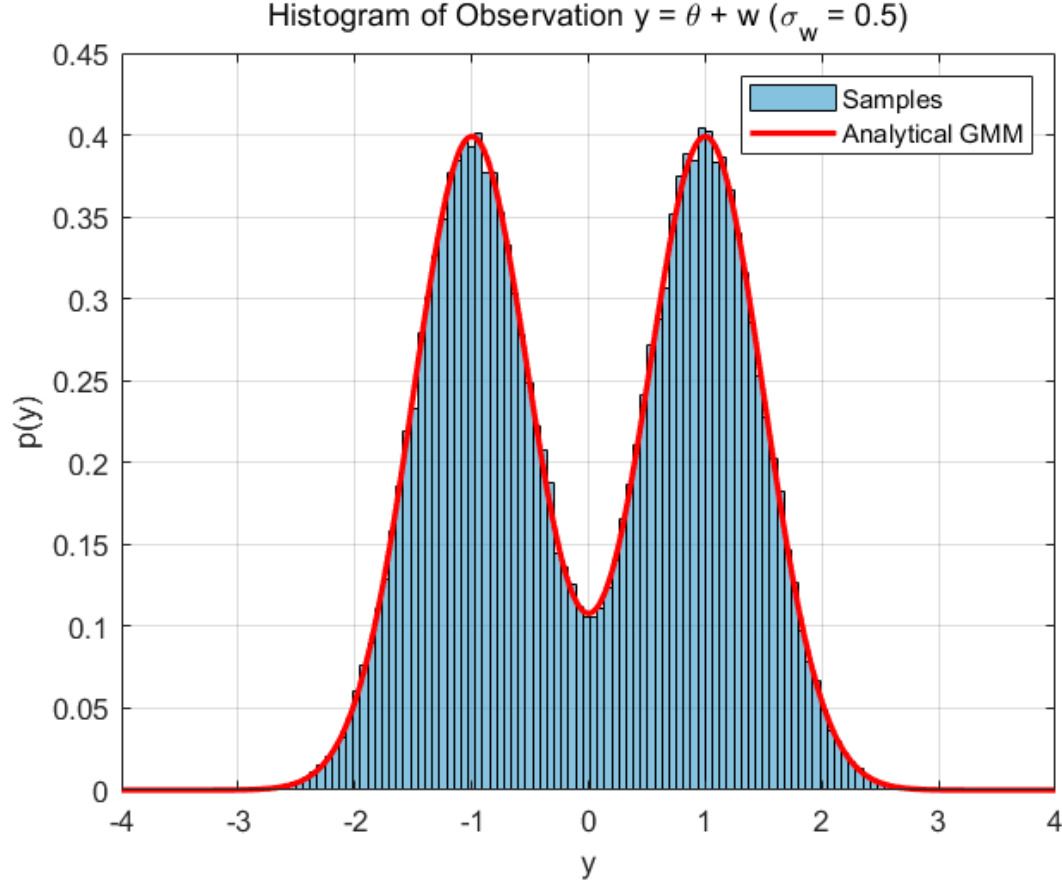


Figure 5: Histogram of Observation $y = \theta + w(\sigma_w = 0.5)$

4.2 (b)

Given a specific observation $y = 0.7$ with $\sigma = 0.5$, we need to make an optimal guess for θ .

My guess for the state is $\hat{\theta} = 1$.

The Maximum A Posteriori (MAP) estimator selects the value of θ that maximizes the posterior probability $p(\theta|y)$. Since the prior $P(\theta)$ is uniform (0.5 for both states), the MAP estimate is equivalent to finding the θ that maximizes the likelihood $p(y|\theta)$. The distance between the observation $y = 0.7$ and $\theta = 1$ is $|0.7 - 1| = 0.3$, while the distance to $\theta = -1$ is $|0.7 - (-1)| = 1.7$. Since 0.7 is significantly closer to the mean of the component centered at 1, the likelihood $p(y|1)$ is much higher than $p(y|-1)$.

Even using the Minimum Mean Square Error (MMSE) estimator, which calculates the conditional expectation $\mathbb{E}[\theta|y]$:

$$\hat{\theta}_{MMSE} = (-1) \cdot P(\theta = -1|y) + (1) \cdot P(\theta = 1|y) \quad (13)$$

At $y = 0.7$, the posterior probability $P(\theta = 1|y)$ is approximately 0.996. Thus, $\hat{\theta}_{MMSE} \approx 0.992$. Both estimators strongly support that $\theta = 1$ is the most plausible underlying state.

4.3 (c)

The observation model is given by $y = \theta + w$. When the value of θ is fixed, it acts as a deterministic constant. Since the noise w is distributed as $\mathcal{N}(0, \sigma^2)$, adding the constant θ simply shifts the mean of the Gaussian distribution. Therefore, the conditional density of y given θ is:

$$p(y|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\theta)^2}{2\sigma^2}} \quad (14)$$

To find the marginal density $p(y)$, we use the Law of Total Probability. Since θ is a discrete random variable, the integral form $p(y) = \int p(y|\theta)p(\theta)d\theta$ simplifies to a weighted sum over all possible values of θ (in this case, $\theta \in \{-1, 1\}$):

$$p(y) = \sum_{\theta_i \in \{-1, 1\}} p(y|\theta_i)P(\theta = \theta_i) \quad (15)$$

Given the prior probabilities $P(\theta = 1) = 0.5$ and $P(\theta = -1) = 0.5$, we substitute these into the equation along with the likelihood derived above:

$$p(y) = P(\theta = 1)p(y|\theta = 1) + P(\theta = -1)p(y|\theta = -1) \quad (16)$$

Substituting the Gaussian PDF for each case:

$$p(y) = 0.5 \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-1)^2}{2\sigma^2}} \right) + 0.5 \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-(-1))^2}{2\sigma^2}} \right) \quad (17)$$

Simplifying the terms, we arrive at the final expression for the marginal density:

$$p(y) = \frac{0.5}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-1)^2}{2\sigma^2}} + \frac{0.5}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y+1)^2}{2\sigma^2}} \quad (18)$$

4.4 (d)

Using Bayes' rule, the posterior probability for the discrete random variable θ given the observation y is expressed as:

$$Pr\{\theta = \theta_i|y\} = \frac{p(y|\theta_i)Pr\{\theta_i\}}{p(y)} \quad (19)$$

1. For $\theta = 1$:

$$Pr\{\theta = 1|y\} = \frac{e^{-\frac{(y-1)^2}{2\sigma^2}}}{e^{-\frac{(y-1)^2}{2\sigma^2}} + e^{-\frac{(y+1)^2}{2\sigma^2}}} \quad (20)$$

2. For $\theta = -1$:

$$Pr\{\theta = -1|y\} = \frac{e^{-\frac{(y+1)^2}{2\sigma^2}}}{e^{-\frac{(y-1)^2}{2\sigma^2}} + e^{-\frac{(y+1)^2}{2\sigma^2}}} \quad (21)$$

These expressions provide a quantitative measure of our belief in each possible state of θ after receiving the noisy measurement y .

4.5 (e)

The Minimum Mean Square Error (MMSE) estimator is defined as the conditional mean of the posterior distribution:

$$\hat{\theta}_{MMSE} = \sum_{\theta} \theta Pr\{\theta|y\} = (1)Pr\{\theta = 1|y\} + (-1)Pr\{\theta = -1|y\} \quad (22)$$

Substituting the posterior expressions and simplifying the exponential terms by expanding the quadratic exponents $(y \pm 1)^2$, the common factor $\exp(-\frac{y^2+1}{2\sigma^2})$ cancels out. This leads to:

$$\hat{\theta}_{MMSE} = \frac{e^{y/\sigma^2} - e^{-y/\sigma^2}}{e^{y/\sigma^2} + e^{-y/\sigma^2}} \quad (23)$$

Using the identity for the hyperbolic tangent function, $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, we obtain the final analytical expression:

$$\hat{\theta}_{MMSE} = \tanh\left(\frac{y}{\sigma^2}\right) \quad (24)$$

This result illustrates that the MMSE estimator provides a "soft" decision that continuously maps the observation y to the range $[-1, 1]$.

4.6 (f)

The Maximum A Posteriori (MAP) estimator is obtained by maximizing the posterior probability with respect to θ :

$$\hat{\theta}_{MAP} = \arg \max_{\theta \in \{-1, 1\}} Pr\{\theta|y\} \quad (25)$$

Since the denominator $p(y)$ is positive and constant for both values of θ , we compare the numerators:

$$\hat{\theta}_{MAP} = 1 \quad \text{if} \quad e^{-\frac{(y-1)^2}{2\sigma^2}} > e^{-\frac{(y+1)^2}{2\sigma^2}} \quad (26)$$

Taking the natural logarithm and simplifying the quadratic terms:

$$-\frac{(y-1)^2}{2\sigma^2} > -\frac{(y+1)^2}{2\sigma^2} \implies (y+1)^2 > (y-1)^2 \implies 4y > 0 \quad (27)$$

This results in a simple and intuitive decision rule. The MAP estimator is the sign of the observation:

$$\hat{\theta}_{MAP} = \text{sign}(y) \quad (28)$$

The comparison highlights a difference in estimation strategies: while the MAP estimator provides a "hard" decision based on the most likely state (sign function), the MMSE estimator provides a "soft" estimate (tanh function) that accounts for the relative confidence in each state based on the noise variance σ^2 .

4.7 (g)

The guess in 4b ($\hat{\theta} = 1$) coincides with the MAP estimator, as it selects the most probable discrete state. In this example, the MAP estimator acts as a "hard" classifier (sign function), while the MMSE estimator provides a "soft" estimate (tanh function) that accounts for uncertainty, especially near $y = 0$.

Table 1: Key Differences between MMSE and MAP Estimators

Feature	MAP Estimator	MMSE Estimator
Formula	$\text{sign}(y)$	$\tanh(y/\sigma^2)$
Decision Type	Hard / Discrete	Soft / Continuous
Criteria	Mode (Most Likely)	Mean (Average)